

Chapter 180

Audio Effects

DaVinci Resolve includes Fairlight FX audio plugins and support for compatible third-party VST and Audio Unit plugins for use in the Edit and Fairlight pages, which let you add effects such as echo, reverb, noise reduction, and aural enhancement to audio clips, tracks, and busses.

This chapter covers different methods of using these effects on clips and tracks.

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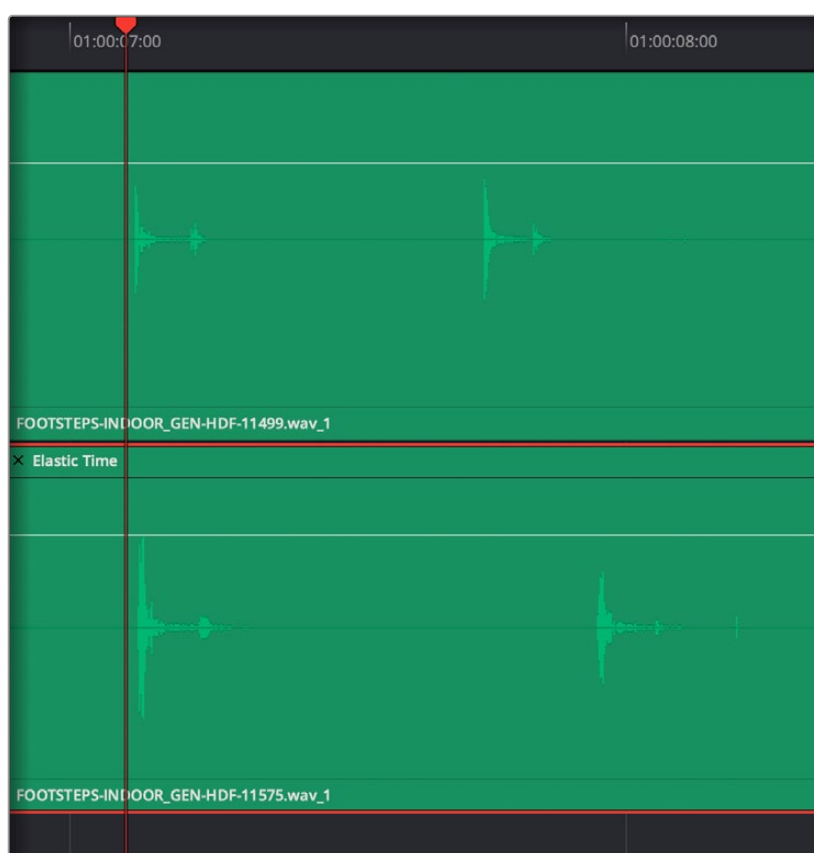
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Elastic Wave Audio Retiming

Elastic Wave retiming is a fast and easy keyframe-based way of dynamically retiming audio, squishing and stretching different parts of a waveform to subtly retime audio playback for a variety of reasons, all while maintaining constant pitch. For example, if you're using the audio from another take to replace that of the current take, but the performer's timing is just a little bit different, you can use Elastic Audio to make small adjustments to retime the second performance to match the first.

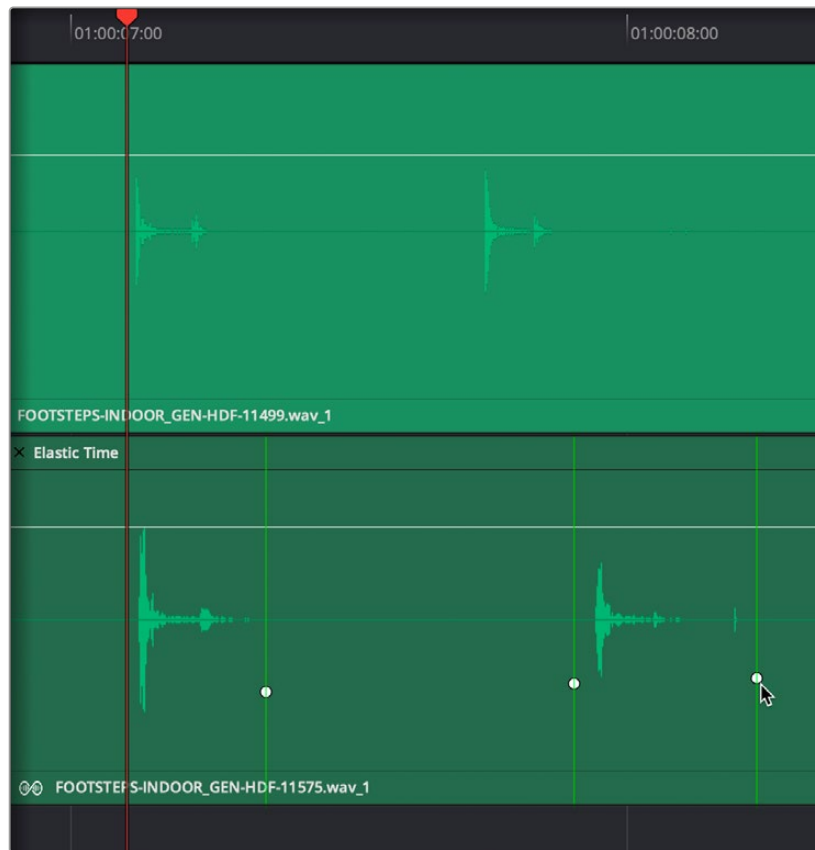
To use Elastic Wave retiming on an audio clip:

- 1 In this example, two tracks of foley effects have been recorded, and the bottom one needs to be retimed to match the top one.
- 2 Right-click an audio clip and choose Elastic Wave retiming from the contextual menu to reveal the Elastic Wave retiming controls.



Enabling Elastic Wave

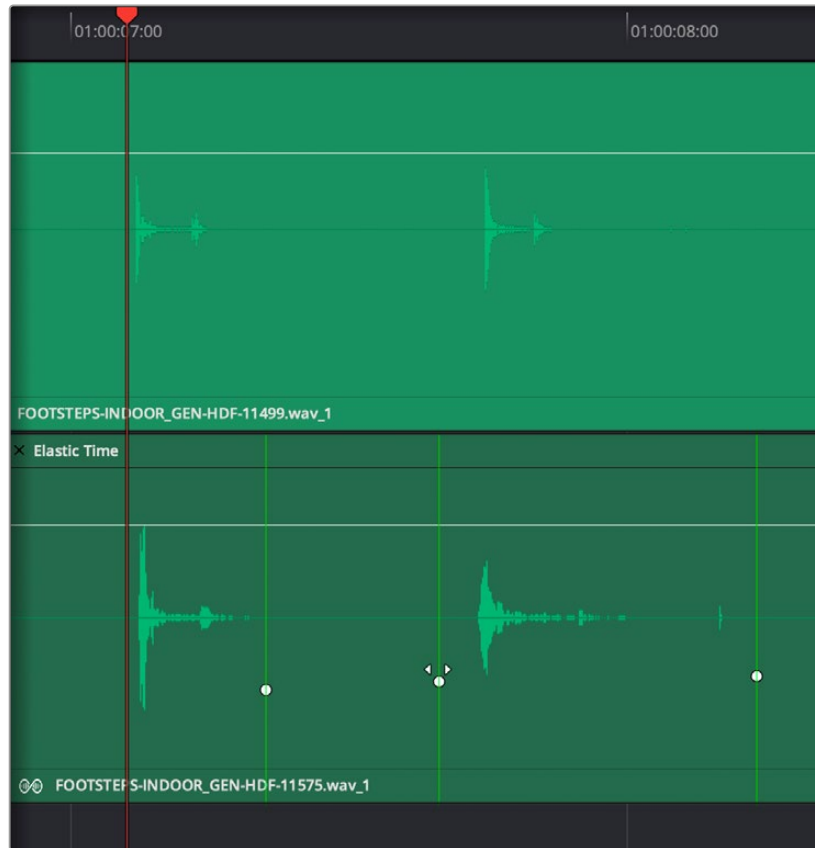
- 3 Command-click anywhere on the clip to add speed keyframes to parts of the waveform you want to retime by stretching or squishing. You can also place speed keyframes to lock parts of a waveform you don't want to retime.



Adding a speed keyframe with Elastic Wave retiming enabled

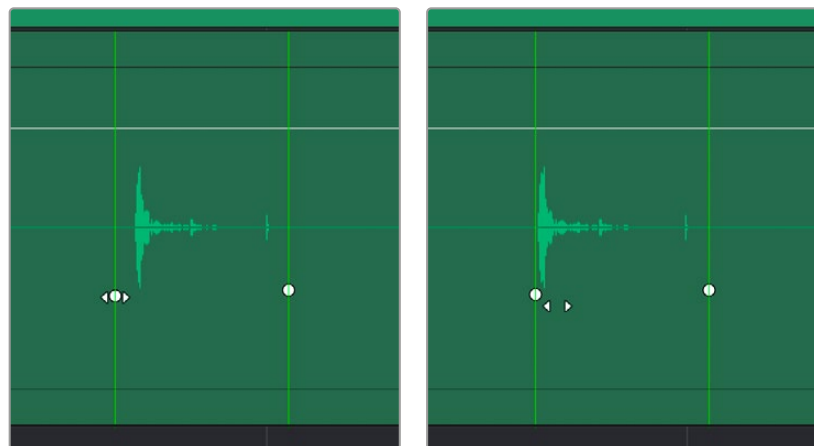
- 4 Dragging the speed keyframe to the left or right speeds up the audio on one side of the keyframe and slows down the audio on the other side of the keyframe, from that keyframe to the neighboring keyframes applied to that clip. Using an audio clip's waveform as your guide, you can use multiple speed keyframes to match the waveform of one performance to the waveform of another, in order to make the timing match. Or, you can adjust speed keyframes freeform to manipulate performances or sounds for creative effect.

You can also drag the beginning or end of an audio clip to retime that portion of the clip going forwards or back to the next speed keyframe that's been added.



Adjusting a series of speed keyframes to retime one performance to match another

- 5 If you've made some speed keyframe adjustments, but you find yourself wishing a speed keyframe you created had been placed at a different position relative to the audio waveform being adjusted, you can hold the Command key down and drag any speed keyframe to move it closer to or farther away from a part of the waveform you want it to retime. This fine-tunes the audio retiming adjustment occurring at that point in the clip.



(Left) Before Command-dragging a speed keyframe to be closer to the original beginning of the sound being retimed, (Right) After

- 6 When you're done, you can click the close button at the upper-left-hand corner to hide the Elastic Wave retiming controls.

To remove Elastic Wave retiming keyframes, do one of the following:

- **To remove a single speed keyframes:** Right-click on a speed keyframe and choose Remove Speed Keyframe from the contextual menu.
- **To remove all speed keyframes and eliminate the Elastic Wave retiming effect:** Right-click a clip and choose Reset Speed Curve.

NOTE: All Elastic Wave retiming adjustments you make in the Fairlight page appear in the Edit page as variable speed effects, accessible using the Retime controls. Be aware that while all Elastic Wave retiming effects can appear as Edit page retime effects, not all Edit page retime effects can appear as Elastic Wave retiming effects on the Fairlight page.

Elastic Wave Options

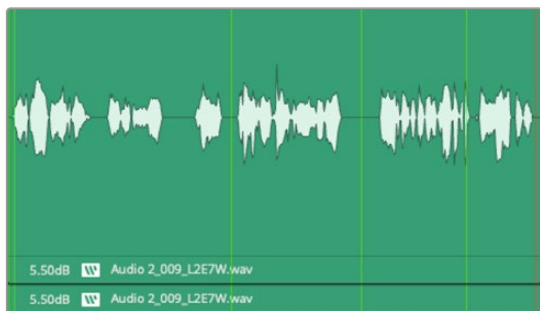
The Elastic Wave context menu has three options for processing:

- **Off:** Elastic Wave is not enabled for the clip (Default).
- **Voice:** Focused on human speech or singing. Note that this is not a good option for other material.
- **General Purpose:** Ideal for music and effects.

Elastic View

Elastic view shows Elastic Wave “time points,” which are similar, but the not the same as, Resolve keyframes. Keyframes represent a point in time plus a value, and Elastic Wave time points represent only a point in time.

When Elastic View is enabled, time points are visible on the selected clip.



Elastic Wave time points



Elastic Wave disabled

When disabled, the time points are not visible, but the info lane still shows that Elastic Wave is enabled with the Elastic Wave icon.

Clearing Timing Points

“Timing points” are the moveable Elastic Wave control points that stretch or compress a given area of a clip. If you choose “Clear Timing Points,” all selected points within a clip are cleared (deleted).

Elastic Wave can operate in the Edit window where video editors may want to re-time spoken word performances so that when the audio is re-timed, the video follows the retiming to match. However, when working in the Fairlight page, re-timing of video is disabled so that you can have separate control of the audio file.

About Audio Plugins

There is no limit to the number of single or multiple plugins when in FlexBus mode that can be applied to clips and to tracks. However there is a limit of 480 when using Fixed Bus Legacy. Plugins are accessed via each track's control strip, or via the Inspector which provides access to both clip and track plugins.

Fairlight FX

Fairlight FX are proprietary DaVinci Resolve-specific plugins that run natively on macOS, Windows, and Linux, providing high-quality audio effects with professional features to all DaVinci Resolve users on all platforms.

Most Fairlight FX plugins, including Channel Dynamics and EQ, and dynamics processors like the Multiband Compressor, Limiter, Reverb, and Ambisonics Meter, are Ambisonics native and offer multichannel processing.

VST and VSTi

VST (Virtual Studio Technology) is an audio plugin standard created by Steinberg. The VST standard allows third-party developers to create plugins for use within VST host applications or create VST host applications.

The Fairlight page supports VST effects from Mono to 7.1 and beyond. These effects can be inserted on mono channels or Link Groups, and compatible VST Ambisonic effects can be used on Ambisonic tracks or busses.

If a stereo VST effect is inserted on an LCR, LCRS, or 5.1 Link Group, the left and right channels will automatically be allocated to the left and right Link Group channels.

DaVinci Resolve supports VST3. VST effects are available on macOS and Windows workstations, but not on Linux.

VST Effects versus VST Instruments (VSTi)

A VST effect is a type of VST plugin that is used to process audio. A VST effect might be a Reverb, Compressor, or EQ. VST Instruments are typically used to synthesize sound or play back sampled audio. VSTs have rapidly replaced hardware synthesizers and dedicated samplers due to their flexibility, repeatability, and low cost.

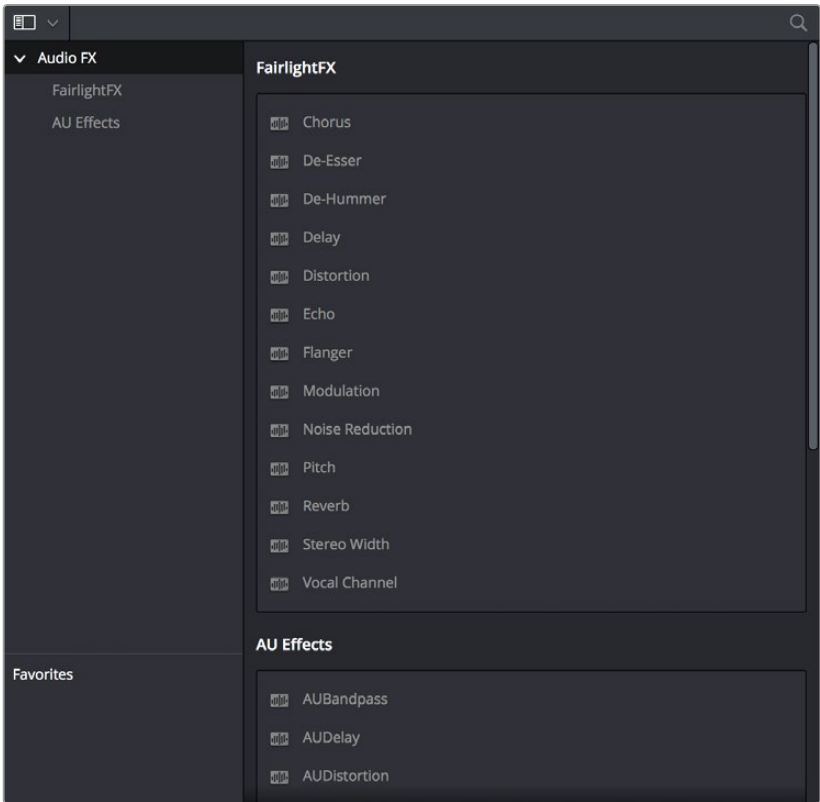
Audio Units

Audio Units is an audio plugin API created by Apple for use on macOS workstations. Like VST, Audio Units (AU) can either process audio or work as instruments.

Compatible AU Ambisonic effects are used on Ambisonic tracks or busses.

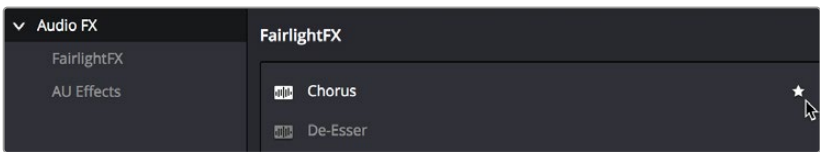
Using Audio Plugins

Fairlight FX are pre-installed with all DaVinci Resolve installations. If you install additional VST or Audio Unit effects on your workstation, they appear in the Audio FX panel of the Effects Library, organized in separate categories.



Audio plugins in the Effects Library

You can click on the far right of any effect to flag it with a star as a favorite effect. When you do so, the favorited effects appear in a separate Favorites area at the bottom of the Effects Library Bin list, and it also appears at the top of the Effects button’s menu on the Mixer when you click the “plus” button.



Stars indicate a flagged favorite effect; all favorites are currently filtered

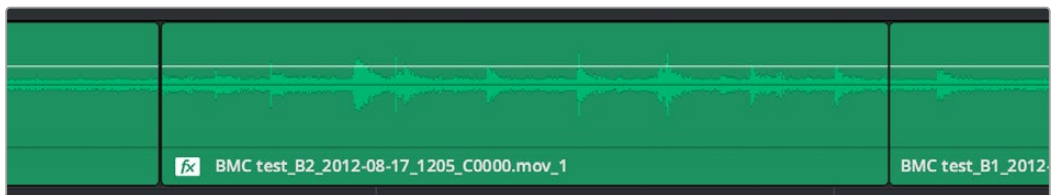
Audio plugins let you apply effects to individual audio clips or entire tracks worth of audio, to add creative qualities such as echo or reverb, or to take care of mastering issues using noise reduction, compression, or EQ.

Applying Plugins

Methods of applying audio plugins to clips on the Fairlight page:

- **To apply an audio effect to a clip:** Drag any effect from the Audio FX panel of the Effects Library onto the clip in the Timeline you want to apply it to.
- **To apply an audio effect to multiple clips:** Select all of the clips you want to apply an audio effect to, then drag any effect from the Audio FX panel of the Effects Library onto any of the selected clips.

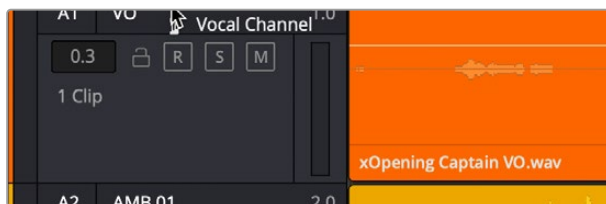
When you apply an audio plugin to a clip, a badge appears to the left of that clip's name bar in the Timeline to let you know there's an effect applied to it.



Timeline clips with audio plugins applied appear with a badge.

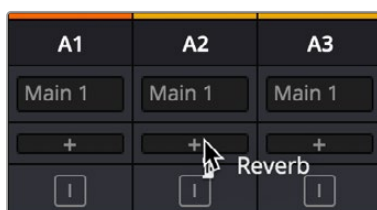
Methods of applying audio plugins to tracks on the Fairlight page:

- **To apply an audio effect to an entire track in the Timeline:** Drag any effect from the Effects Library onto the track header.



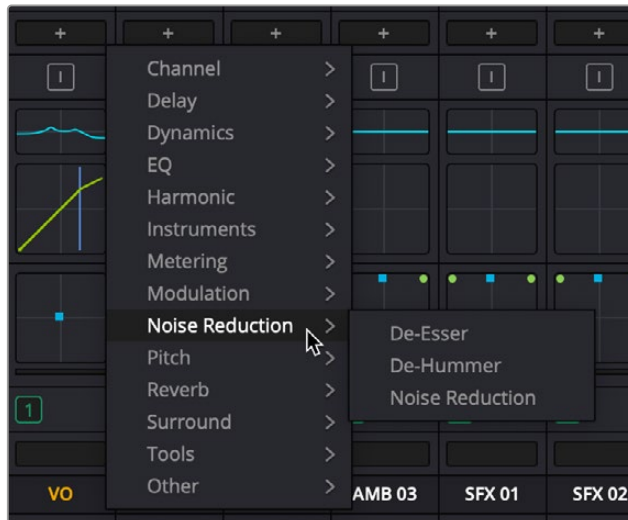
Applying an audio effect to a whole track via the Timeline

- **To apply an audio effect to a track or bus in the Mixer:** Drag any effect from the Audio FX panel of the Effects Library onto the clip in the Timeline you want to apply it to.



Applying an audio effect to a whole track via drag and drop to the Mixer

- **To apply an audio effect to a track or bus using the Mixer controls:** Click the plus button in the channel strip of the track you want to apply an effect to, and then choose an effect from the dropdown menu that appears. All filters appear within categories to make them easier to find. If you've clicked the star button of any filters in the Effects Library to favorite them, these favorite filters appear at the top of the plus button's dropdown list.
- **To copy audio effects from one mixer channel to another:** Hold the Option key and click and drag the effect to the desired channel and slot to copy.
- **To reorder audio effects in the Mixer:** Click and drag the audio effect to move to the desired slot position.



Applying an audio effect to a whole track via the Mixer's own controls

TIP: The categories that VST and Audio Units effects are organized into in the Mixer can be edited in the Audio Plugins panel of the DaVinci Resolve System Preferences window. All plugins on a workstation are shown in the Available Plugins list, and clicking within the Category column lets you use a dropdown menu with which to change categories.

NOTE: This does not apply to Fairlight FX plugins.

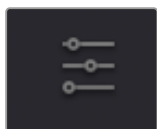
Editing Plugin Parameters

To edit a clip's audio plugins:

- Select that clip and open the Inspector. All audio plugins applied to that clip appear in the Effects tab, with that effect's controls appearing directly in the Inspector.

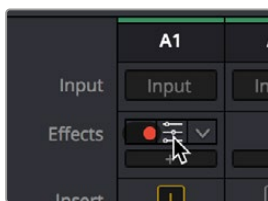
To edit a track's audio plugins, do one of the following:

- Click in the background of the Timeline header to select that track, and then open the Inspector. Click the custom UI button for that effect to open its controls.



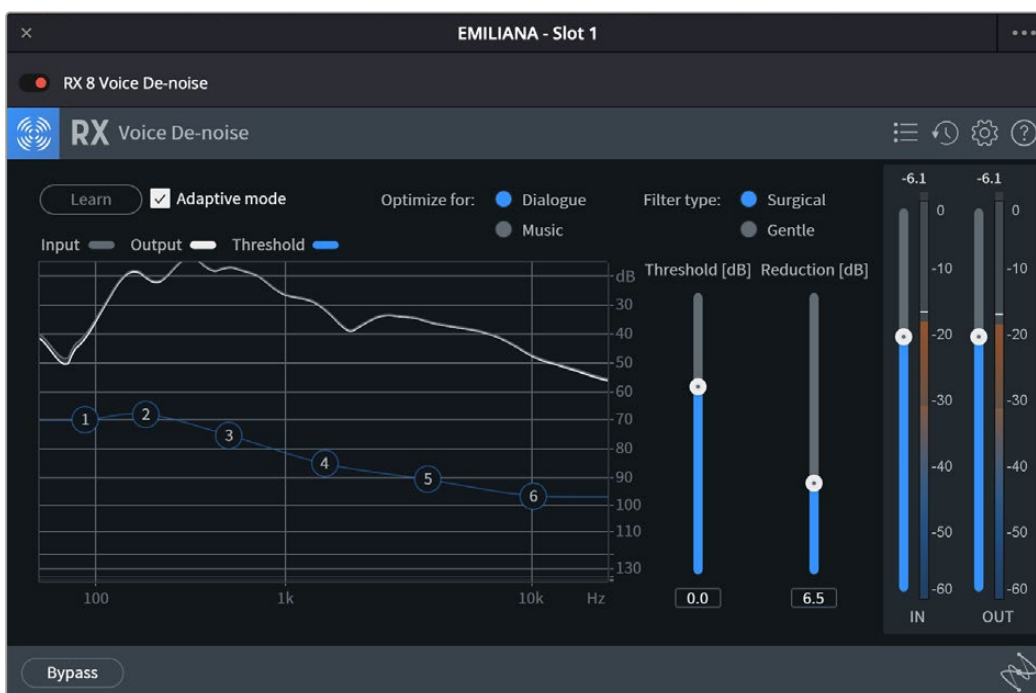
The custom UI button for audio plugins in the Inspector

- Move the pointer over the plugin's name in the Effects area of the Mixer, and click on the custom UI button to open its controls.



The custom UI button for audio plugins in the Mixer

Nearly all Fairlight FX, VST, and Audio Unit audio plugins have a custom user interface that makes it much easier to manipulate that effect's controls. These can be opened from within DaVinci Resolve.



The custom audio effect interface for iZotope RX Voice De-noise

Working with Plugins in the Inspector

Methods of working with audio plugins in the Inspector:

- **To rearrange the order of multiple audio effects applied to a clip:** Click the move up or move down buttons in any effect's title bar, to the left of each effect's Trash Can button.
- **To disable or re-enable an effect:** Click the toggle button at the far left of each effect's title bar.
- **To remove an effect:** Click the Trash Can button.
- **To reset any effect parameter:** Click the Reset button at the far right of the parameter you want to reset.
- **To open or collapse an effect's parameters:** Double-click the title bar.
- **To open or collapse the parameters of all effects:** Hold the Option key down and double-click any effect's title bar.

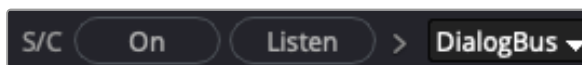
Once applied to a clip or track, audio plugins can also be keyframed or automated just like volume and pan settings, to create dynamic audio effects that change over time.

Sidechaining

DaVinci Resolve supports the sidechaining of compatible Fairlight FX, AU, and VST plugins, which lets you temporarily lower the level of a target track or bus using the audible signal from a source track or bus.

This process, also known as “ducking,” is helpful when multiple tracks compete for space in a mix because they share some or all the same frequencies, such as a track or bus with dialogue and a bus with a crowd cheering at a sports event.

In the example above, the dialogue is fed to the sidechain input of a compatible dynamics plugin (i.e., a compressor or limiter) on the Mixer channel for the crowd noise. When the controls on the dynamics are set correctly, the voices on the Dialogue channel will subtly and slightly lower or attenuate the level of the crowd noise, allowing the voices to be heard more prominently in the mix. When the dialogue stops, the cheering returns to its original level.



Sidechain controls

To use sidechaining

- Open the compressor or limiter on the Mixer channel with the crowd noise.
- Select the Dialogue channel in the source dropdown at the top of the plugin, to the right of the Listen button.

This is the audio you’re using to attenuate the level of the cheering crowd.

Using a bus as the ducking source lets you use multiple sources to trigger changes in your target track.

- Click the On button, which turns red, indicating that the Sidechain (S/C) function is active.
- Adjust the controls to taste during playback of your mix, while ensuring the effect sounds natural.

You can experiment with this procedure using the Compressor section of the Channel Dynamics plugin or the Fairlight FX Limiter.

For more information on sidechaining with the Channel Dynamics plugin, see *Chapter 177, “Mixing in the Fairlight Page.”*

For more information on sidechaining with the Fairlight FX Limiter, see *Chapter 181, “Fairlight FX.”*

- Clicking the Listen button lets you hear the incoming signal if you need to confirm that the correct input signal is coming into the Limiter. This button turns yellow when active.

Applying Audio Plugins to Busses

You can apply audio plugins to Main and Sub buses just like any other track, with which to apply any audio mastering effects operations you require to individual submixes, or even to an entire main.

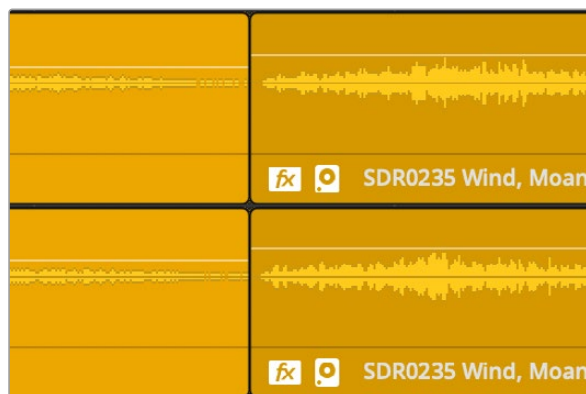
TIP: You can quickly assign plugins for a selected group of channel strips, or to all channel strips, by holding down the Option key (Mac) or the Alt key (Windows) for all selected tracks or Command-Option (Mac) or Control-Alt (Windows) for all mixer channel strips prior to performing the operation. These shortcuts can save a lot of time in your workflow.

Dealing With Processor Intensive Plugins

As you apply more and more plugins directly to clips in complicated mixes, you may discover you lack the processing power to play all audio tracks and effects in real time. When this happens, there are two ways you can ease the burden that audio clip effects are placing on your workstation.

Caching Audio Clips With Plugins

One quick fix is to manually enable the caching of one or more selected audio clips with audio effects applied to them to improve your project's performance. Once an audio clip is cached, all plugin effects are "baked in" and that clip's audio waveform updates to reflect the altered audio. Cached clips appear with a small badge to the right of the FX badge in the name bar of the clip.



A cache badge lets you know this audio clip has cached effects

This is a non-destructive operation that has no lasting effect on the source media of cached clips. You can still alter a cached clip's plugin parameters whenever you want. Opening the graphical controls of a cached clip temporarily suspends audio caching, and when you're finished, the clip is automatically re-cached and its waveform is updated to reflect the change so long as Cache Audio Effects remains enabled.

To cache audio effects for one or more selected clips:

- Right-click an audio clip with a plugin applied and choose Cache Audio Effects to enable audio effect caching for that clip. If you right-click one of multiple selected clips, you'll enable Cache Audio Effects for all selected clips at once. Once enabled, that clip will be continue to be cached (and re-cached if you change the plugin's parameters) until you manually disable audio caching.

To disable audio caching for one or more selected clips:

- Right-click an audio clip that's been cached and choose Cache Audio Effects to disable audio effect caching for that clip.

Exporting Audio Clips With Plugins

Another way of easing the burden of audio clip effects on your system is to export the effected clip to another layer. This creates a new piece of audio media with the effect “baked in,” which is written to the directory location specified by the “Save clips to” field of the Capture and Playback panel of the Project Settings.

To bounce one or more selected audio clips with effects to another layer:

- Right-click an audio clip with a plugin applied and choose Export Audio Files. If you right-click one of multiple selected clips, you'll have options for what you would like exported for each selected clip all at once.

Exported clips no longer have editable effects, but you can always choose View > Show Audio Track Layers to see the original underlying clip that still has the original plugin effect applied, and you can unmute it, move it back up to the top, edit the effect, and export another version of the clip, which appears as the topmost layer.

For more information on exporting clips to files, see *Chapter 175, “Editing Basics in the Fairlight Page.”*